

animals, BreedCon for indigenous cow/buffalo breed registry, and www.pashubazar.com, a site where animals and related products and accessories can be bought and sold. Pashubazar is being implemented with the help of village dairy co-operatives, veterinary hospitals and rural agri-business centres. This is probably the first ever effort towards using information technology in the livestock sector in rural areas.

As a benchmark for quality, the brand Chitale is able to command premium pricing for most of its products. Even a litre of Chitale milk—perhaps the most generic commodity in the value chain—sells for a good rupee more per litre than that from competitors.

The Chitale brand equity is an outcome of careful attention to many details, including a feeding system that ensures that every buffalo consumes an exactly pre-determined portion of feed. The monitoring system at the farm in Bhilwadi includes two tags around the neck of a buffalo, an orange ID tag and a blue smart tag. The former is a simple numbering associated with the ID stored in a computer, and the blue tag is a metallic cover under which a smart card has been locked. Buffaloes aren't really expected to maintain discipline, hence the metallic cover—which ensures there's no damage to the card inside.

Every time a buffalo enters the milking parlour, the transponder—a device that transmits and responds, hence the word—in the strap is identified. A warning lamp lights up on the milking point controller (MPC) at the appointed symbol, if the buffalo has not been milked, or if the milk has not been directed to the tank, as in the case of theft. If the buffalo does not produce as much as expected, a different warning light glows.

Manning The Herd

It was at this level of existing automation that Dr Samad began his pilot test of Herdman. According to him, the software has been developed for the small- to medium-scale farmer with a minimum of 50 animals. Created in 2003 on a VB-Access configuration, the present version of Herdman uses .NET for the front-end and SQL Server for the back-end.

Every animal in the herd is registered based on its unique ID, and herds and lots are created. Next is the various data entry modules. Records are maintained about pregnancies, calving, illness, treatment, milking, and whether a particular animal is in oestrus.

Periodic monitoring of these records, according to Dr Samad, takes the focus away from “fire-fighting” to health management. For example, a periodic reading of the milking records identifies low performance when it occurs—even a variation as low as 500 ml a day. Herdman not only calculates the averages for the herd or lot, but also the standard deviation for the herd: a low standard deviation indicates that the problem has a common source which needs to be identified, whereas a high standard deviation indicates that only a few animals in the herd are low performers. Effective use of the software does not require daily data entry for milk; once-in-a-month figures are all that the International Committee on Animal Record stipulates. Other data such as insemina-

tion, pregnancy, calving and treatment is recorded as it occurs in the farm or village.

Statistics And Decisions

Dr Samad further explains how Herdman helps in right decision-making.

When lactation is plotted against days, the curve must peak at 30 to 40 days. Also, 40 per cent of the total productivity is to be achieved in the first 100 days. If this does not happen, the cause of the problem is considered from the earlier-mentioned statistical technique.

In the event of it being a “common-source problem,” the next step is to identify that common source. A typical correlation is that the feed quality or quantity is not proper. This is when Herdman's “Metabolic Profiling” module comes into play: the program identifies “random samples” from the entire herd. This has to include a weighted sample of animals from each category: freshly-calved, mid-lactation, late lactation and dry.

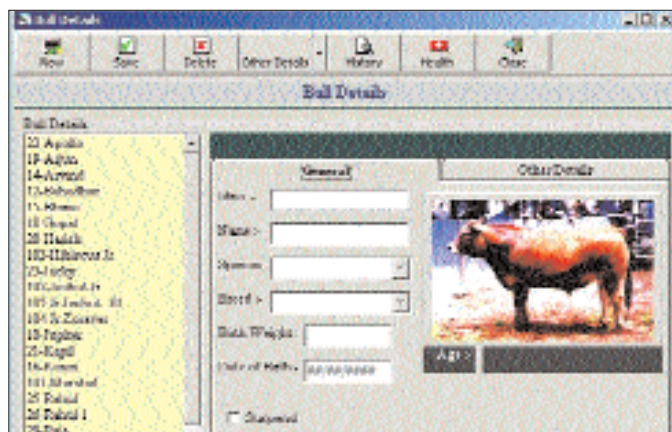
Blood samples are taken from each, and the results are analysed by Herdman as mean plus or minus standard deviation for different haematological parameters—such as RBC count—and chemistry parameters. For feed decisions, chemistry parameters such as blood glucose, blood urea, serum proteins, calcium and other minerals are considered.

The Software

Other features of Herdman include:

Action List: Farm activity scheduling is enabled with an “Action” list generated from the database. List of animals in oestrus (“heat”), those eligible for pregnancy examination, drying off, due for calving, milk recording, and so on, are generated. This streamlines the farm management practice, which no more depends on mere observation.

Alarm List: A list of animals not performing in accordance with standards of production and reproduction.



Every animal can be monitored using Herdman

Feed Formulation and Scheduling: Based on the feed requirement for different categories of animals—lactating, dry, pregnant, and so on—Herdman generates a feeding schedule. Cost of feed and cost per animal, too, can be tracked.

Herd Performance Indices: Production and reproduction-based herd indices are calculated as mean plus or minus standard deviation for each